

Supporting Information:

Supporting Information: Objective-Dependent Active Learning and Calibrated Uncertainty for Sample-Efficient Discovery of Rare-Earth Extractants

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Datasets

***f*-element extraction (primary).** 1,202 experimental lanthanide distribution coefficients ($\log D$) from Liu et al. (1,085 training + 117 validation), with 2,291 features per record: ligand ECFP/Morgan bits and RDKit physicochemical descriptors, experimental conditions (ligand and acid concentration, diluent identity and volume ratio, temperature), and lanthanide-identity descriptors. Target range $\log D \in [-4.00, 3.59]$. A strictly prospective external set of four subsequently synthesised ligands (56 ligand $\times$ metal measurements) is used only for final validation.

Lipophilicity (generality). 4,200 octanol–water $\log D$ values (MoleculeNet). Features: 1,024-bit Morgan fingerprints (radius 2) concatenated with the RDKit physicochemical descriptors above.

Methods

Surrogate. Random forest, 300 trees, minimum 2 samples per leaf, features standardised. Epistemic uncertainty $\hat{\sigma}$ is the per-point standard deviation across trees.

Conformal calibration. Split-conformal: 20% of the training data held out as a calibration set; nonconformity scores $s_i = |y_i - \hat{\mu}_i|/\hat{\sigma}_i$; the calibrated interval half-width is $q_\alpha \hat{\sigma}$ with q_α the empirical α -quantile of $\{s_i\}$.

Active-learning protocol. Fixed 80/20 pool/test split. f -element: seed = 25, batch = 15, budget = 420. Lipophilicity: seed = 40, batch = 25, budget = 715. Six random seeds per strategy; curves report mean $\pm$ s.d. Top-10 recall is the fraction of the ten highest-log D pool members acquired.

Full results

Table S1: Active learning on the f -element log D benchmark (pool 962, test 240). R^2 at fixed budget; final R^2 and top-10 recall at budget 430; measurements to reach $R^2=0.85$.

strategy	$R^2@100$	$R^2@250$	final R^2	recall	labels $\rightarrow$ 0.85
random	0.543	0.754	0.851	0.50	404
greedy	0.272	0.506	0.633	1.00	n/a
uncertainty	0.396	0.679	0.870	0.38	400
UCB	0.438	0.600	0.701	1.00	n/a
EI	0.379	0.664	0.776	0.98	n/a
hybrid	0.379	n/a	0.782	0.93	n/a

Baseline and prospective validation

A random forest trained on the published training split attains $R^2 = 0.942$, RMSE= 0.328 on the validation split (cf. the reported deep network $R^2 = 0.85$, RMSE= 0.53). Trained on all 1,202 measurements and applied to the four prospective ligands, it attains $R^2 = 0.687$, RMSE= 0.639, MAE= 0.502 log units.

Table S2: Active learning on the Lipophilicity benchmark (pool 3360, test 840), Morgan-fingerprint features. Final values at budget 715.

strategy	$R^2@150$	final R^2	recall
random	0.253	0.497	0.27
greedy	0.105	0.206	0.53
uncertainty	0.212	0.501	0.20
UCB	0.139	0.281	0.57
EI	0.143	0.309	0.53
hybrid	0.165	0.384	0.30

Table S3: Uncertainty calibration (f -element benchmark, 10 seeds). Empirical coverage of nominal- α intervals.

nominal α	uncalibrated	conformal
0.50	0.656	0.503
0.80	0.904	0.778
0.90	0.950	0.886
mean calib. error	0.103	0.013

Reproduction

All code is released (concept DOI 10.5281/zenodo.20764583). The experiments are reproduced by `src/05_active_learning/`: `real_data.py` (baseline), `real_al.py` (f -element AL), `lipo_al.py` (Lipophilicity AL), `al_calibration.py` (calibration), `prospective.py` (prospective validation). The f -element source data are from Liu et al. (cited); the Lipophilicity set is from MoleculeNet.

Table S4: Active learning with vs. without conformal calibration (final R^2 / top-10 recall). Acquisition performance is unchanged within noise.

strategy	uncalibrated	conformal
UCB	0.647 / 1.00	0.635 / 1.00
EI	0.696 / 0.98	0.709 / 0.98
hybrid	0.706 / 0.90	0.706 / 0.90